

WASP-12b

R-1002

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_161231 · created 2026-07-30T01:35:31+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457752.67718 +/- 0.00035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.045
Transit depth (Rp/Rs)^2	2.34 +/- 1.37 [%]
Orbital Inclination (inc)	77.25 +/- 2.93
Airmass coefficient 1 (a1)	2.0 +/- 1.53
Airmass coefficient 2 (a2)	-0.64 +/- 0.29
Scatter in the residuals of the lightcurve fit is	82.12 %
Best Comparison Star	#2 - [301, 399]
Optimal Aperture	8.54
Optimal Annulus	15.65
Transit Duration (day)	0.109 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.045 vs 0.1178 (deviation 0.53σ)
Duration fit vs published	0.109 d vs 0.125 d (deviation 0.64σ)
Mid-time O–C	-1571.0 min
Depth SNR	2.3
Residual scatter	82.12 %

Light curve

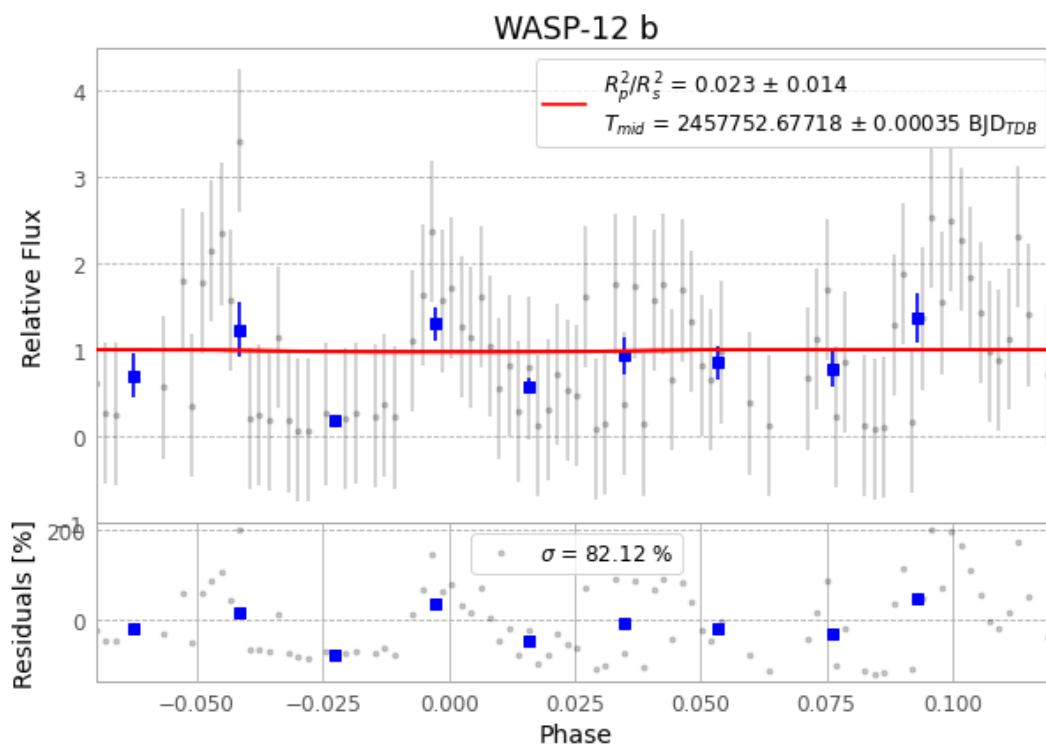


Figure 1 — FinalLightCurve_WASP-12 b_2016-12-29.png (SHA-256 2c6e3d3a22e44af1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [386, 431], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 97ef85854042ffc2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

103 FITS frames (68 MB), WASP-12161230020912.FITS → WASP-12161230071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-161231.log (6d69c004fff3...), FinalLightCurve_WASP-12 b_2016-12-29.png (2c6e3d3a22e4...), FinalLightCurve_WASP-12 b_2016-12-29.csv (ae825f97c3fb...)

Compute resources

Wall time 265.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 01:35Z.